

Hadoop - Theory

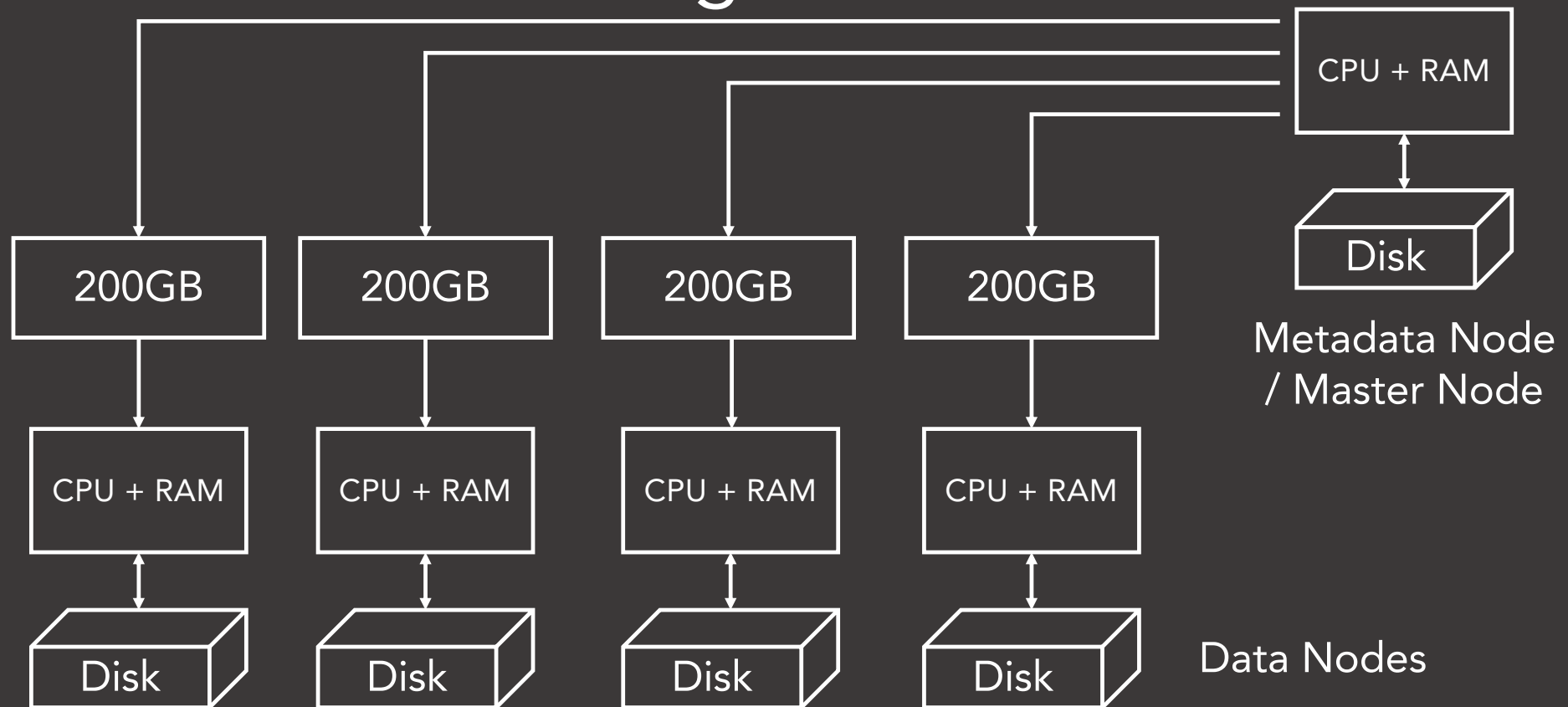
Agenda

- What is Hadoop
- MapReduce
- HDFS
- How MapReduce works
- YARN
- Ecosystem Built around Hadoop

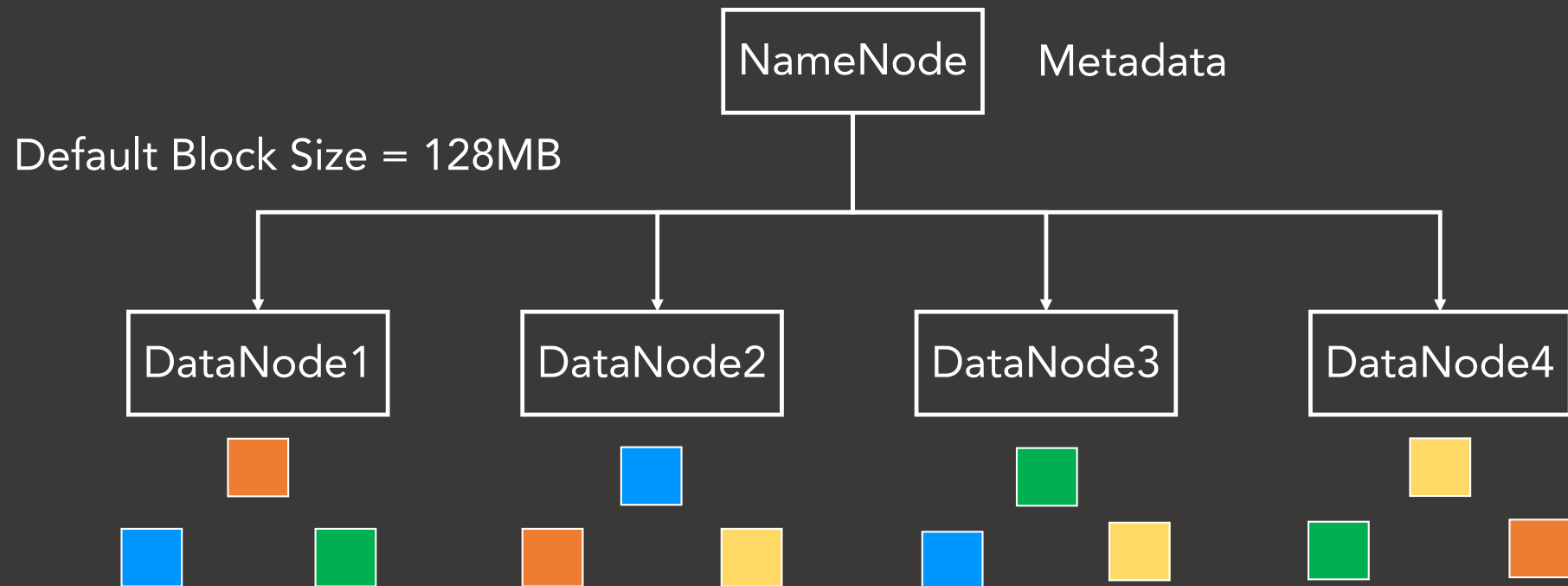
What is Hadoop

- Hadoop is a framework that is used to store and process large datasets.
- HDFS – Distributed Storage
- MapReduce – Distributed Compute

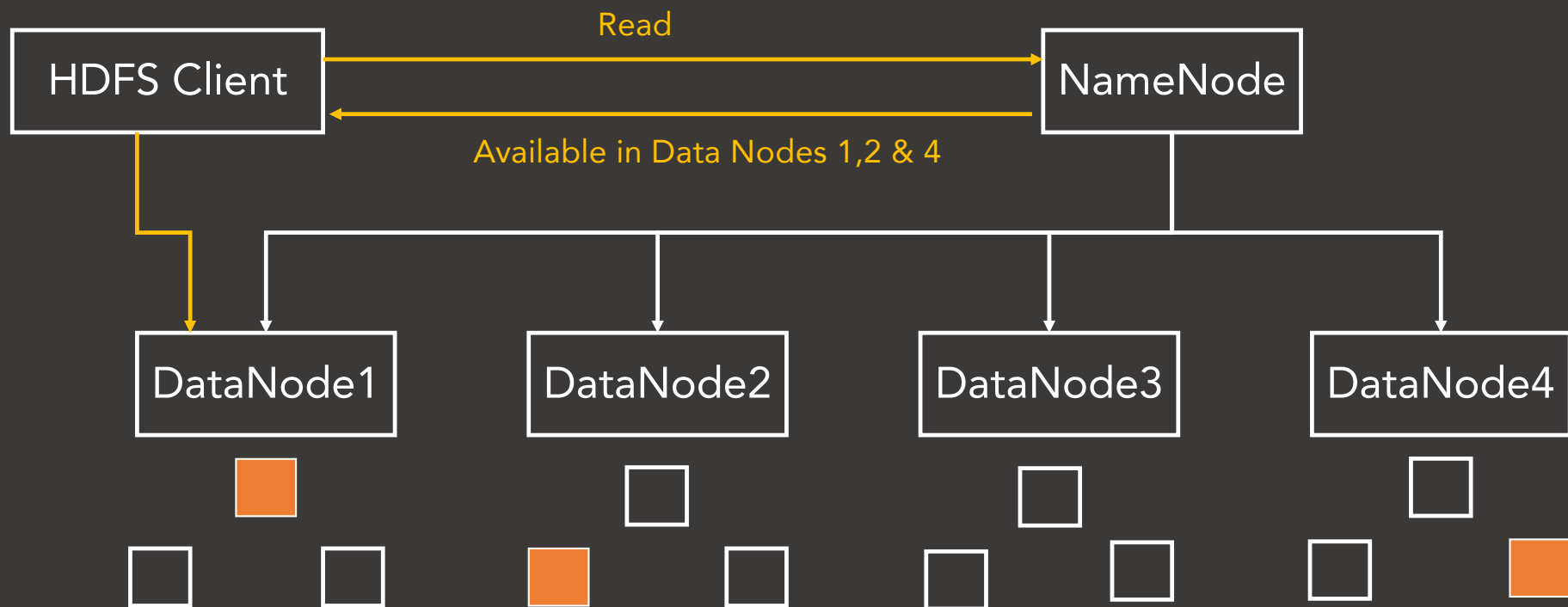
Distributed Storage



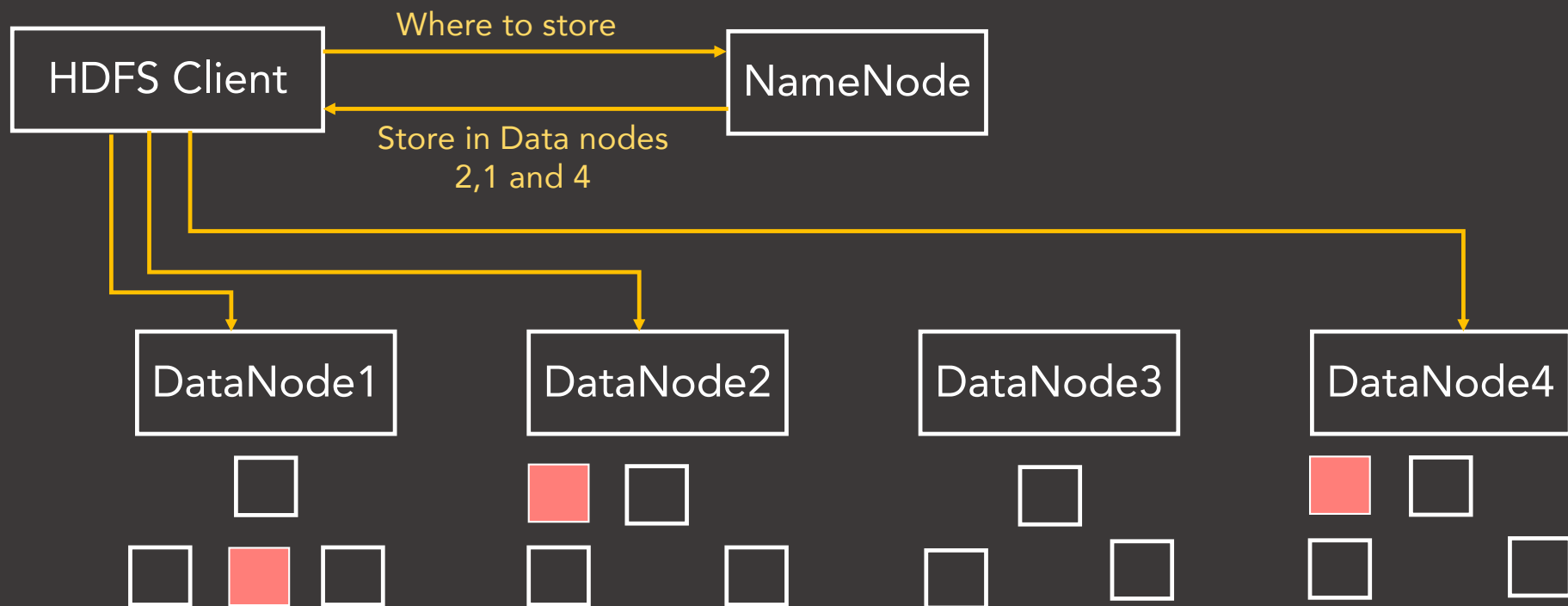
HDFS



HDFS – Read Path



HDFS – Write Path

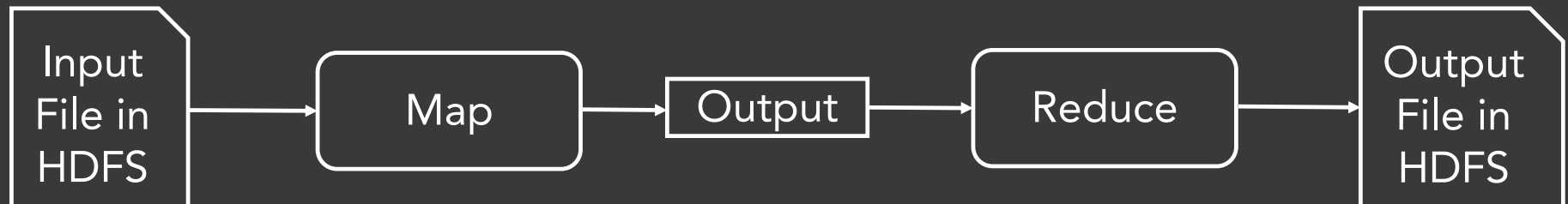


MapReduce

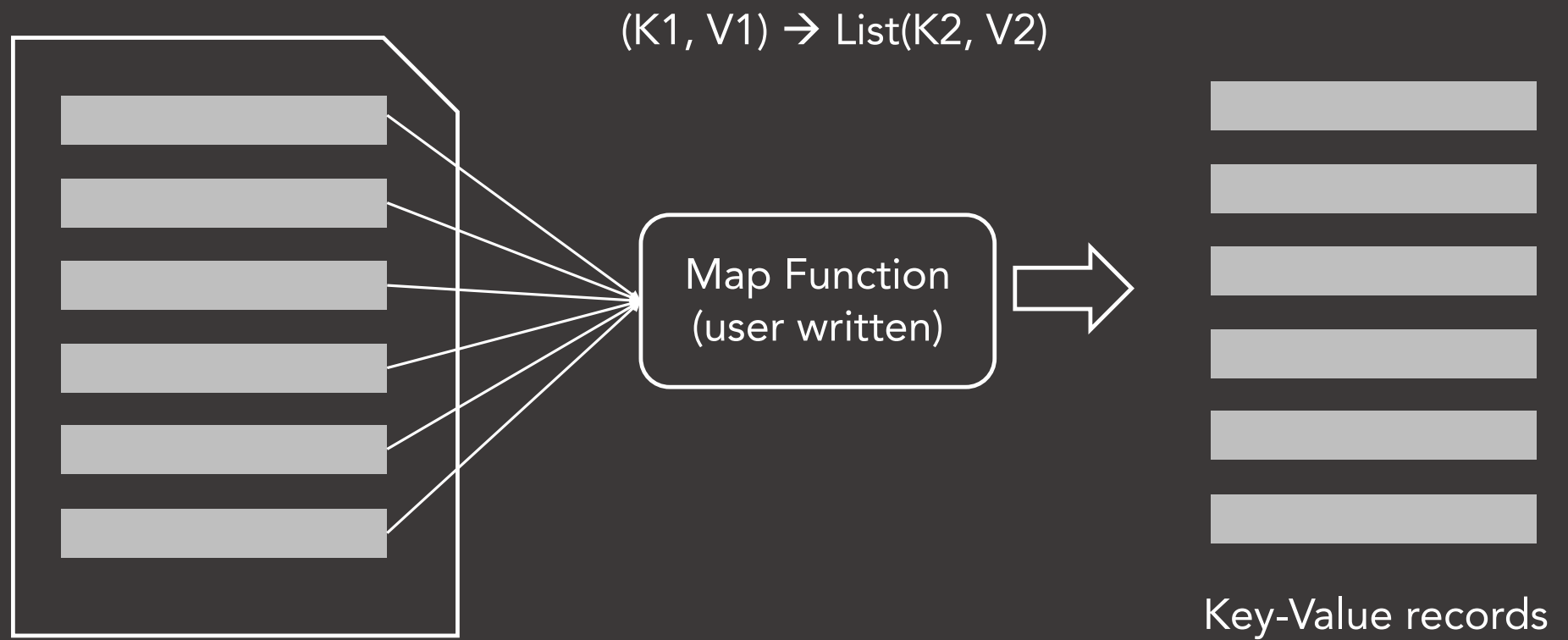
- Programming model/framework that does distributed processing of (distributed) data.
- Programming languages – **Java**, Python, Ruby.
- The framework does a lot of work for us.
- Two main phases:
 - Map
 - Reduce

Map and Reduce

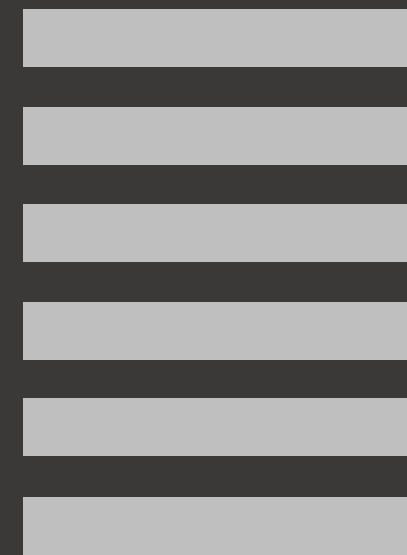
- Processing is done in two phases – Map and Reduce.
- Map function (written by developer).
- Reduce function (written by developer).
- Input to Map phase is the raw file.
- Input to Reduce phase is output of Map.



Map

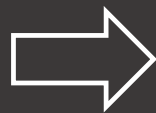


Reduce



Key-Value records

$(K2, \text{List}(V2)) \rightarrow \text{List}(K2, V3)$

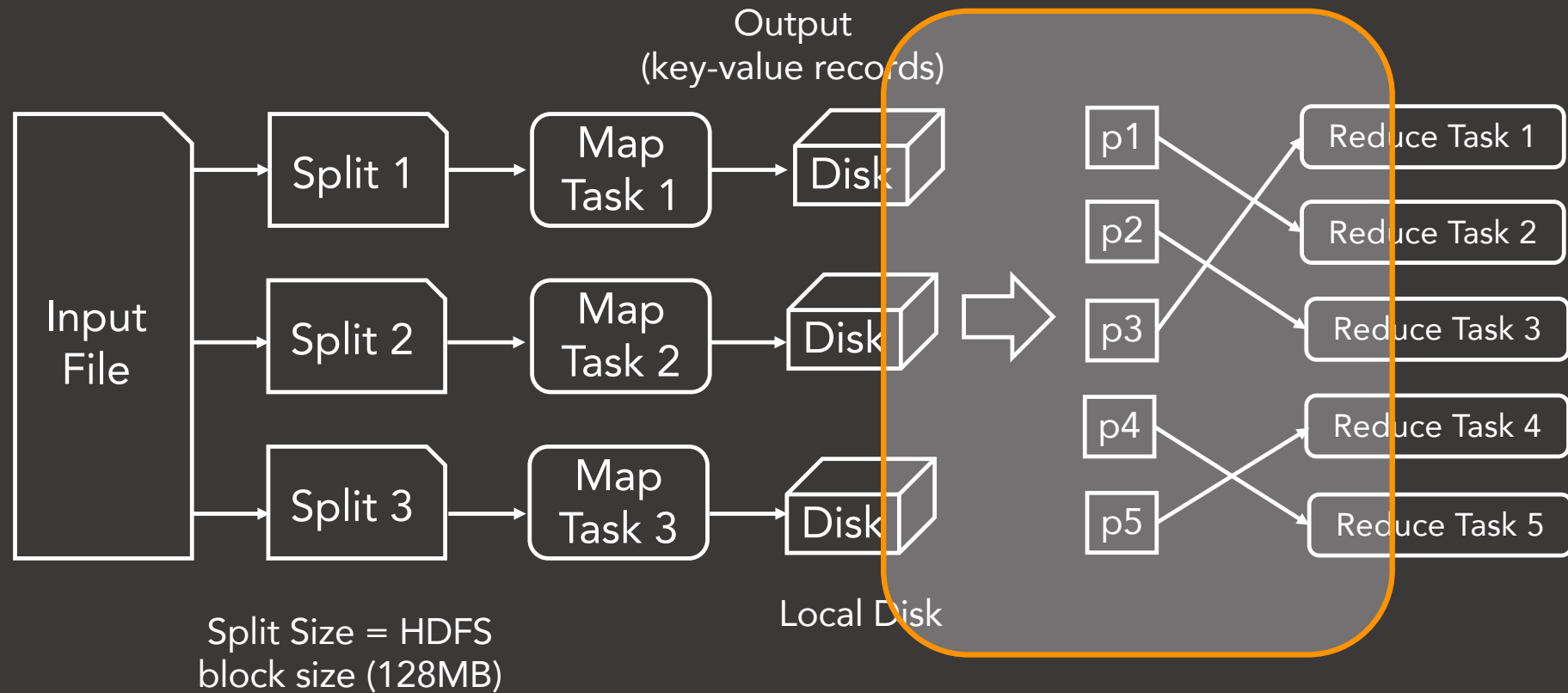


Reduce Function
(user written)

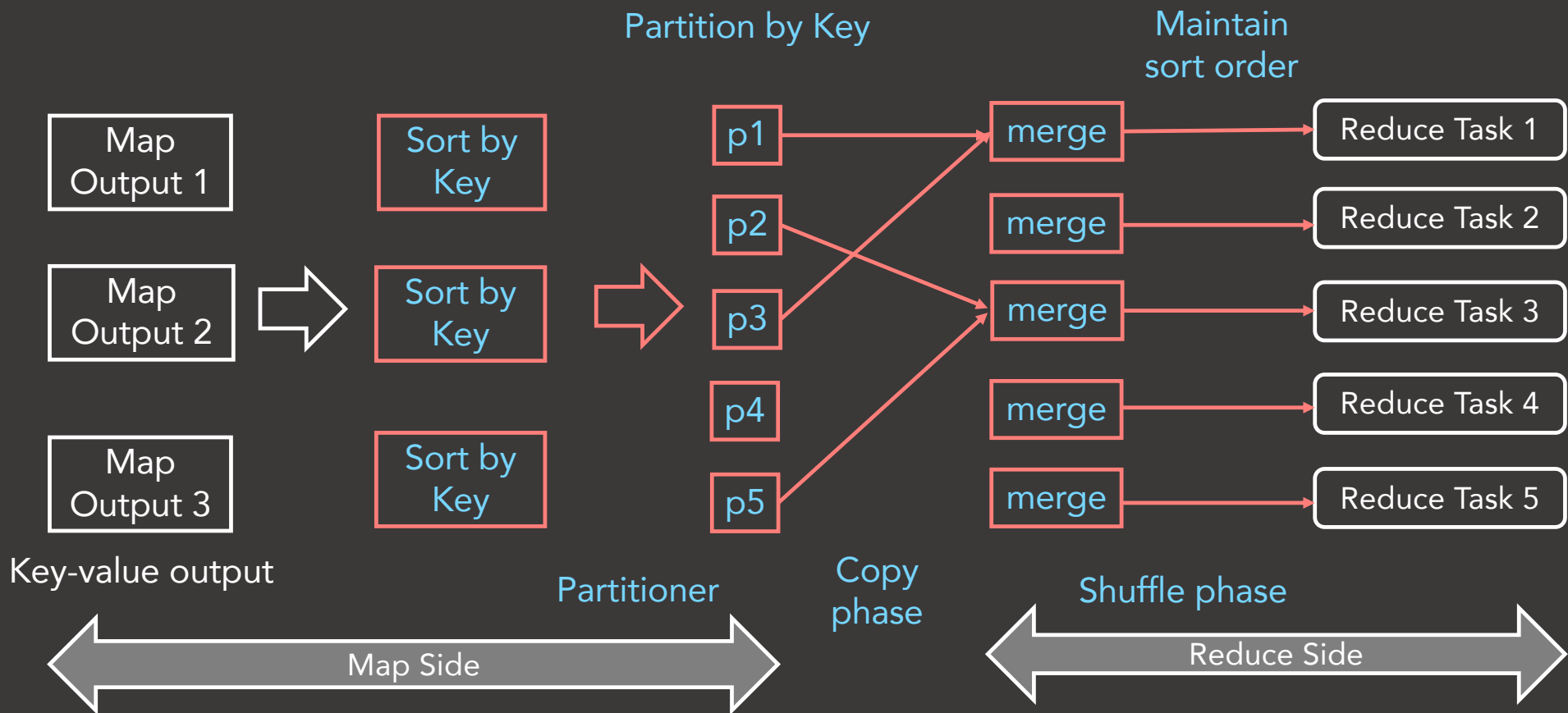


Aggregated
Output

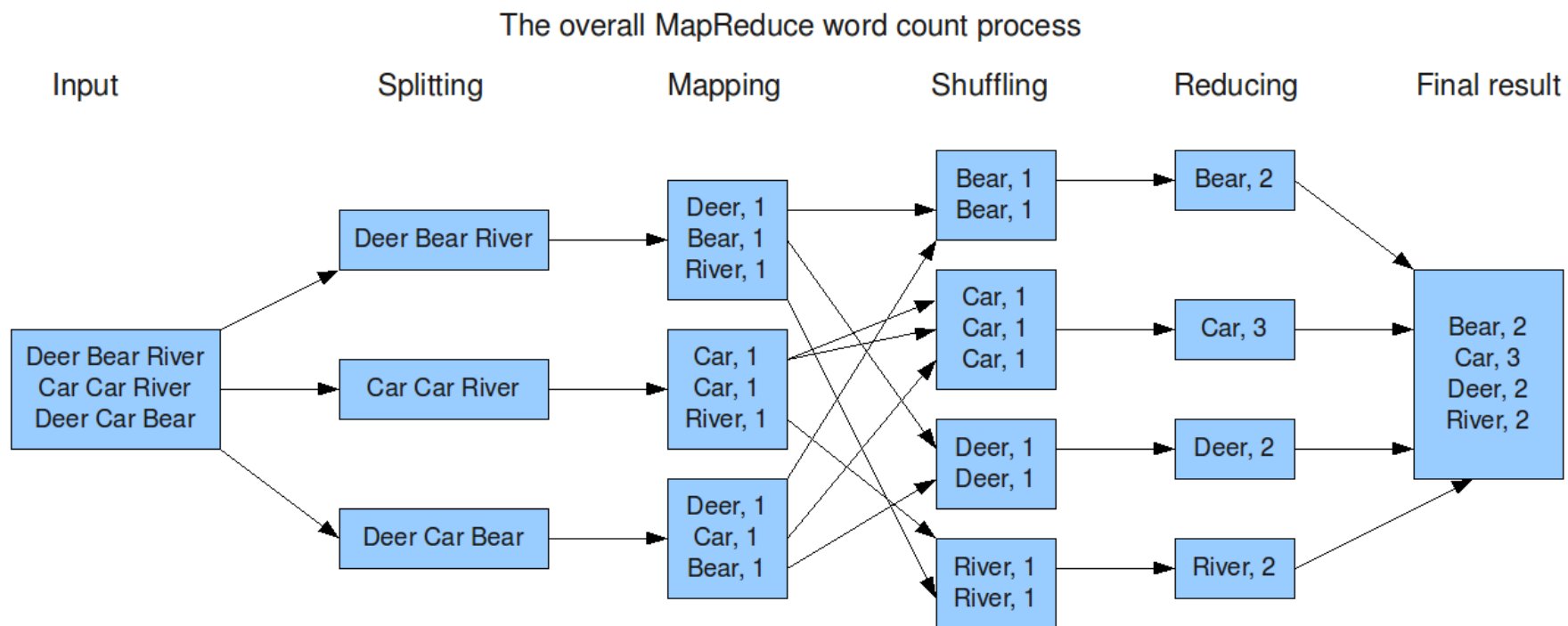
How MapReduce Works



How MapReduce Works



How MapReduce Works - Example

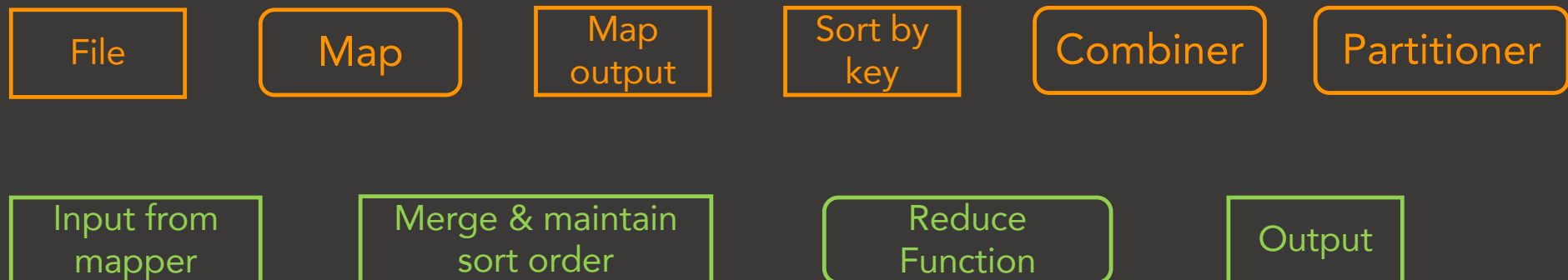


Partitioner

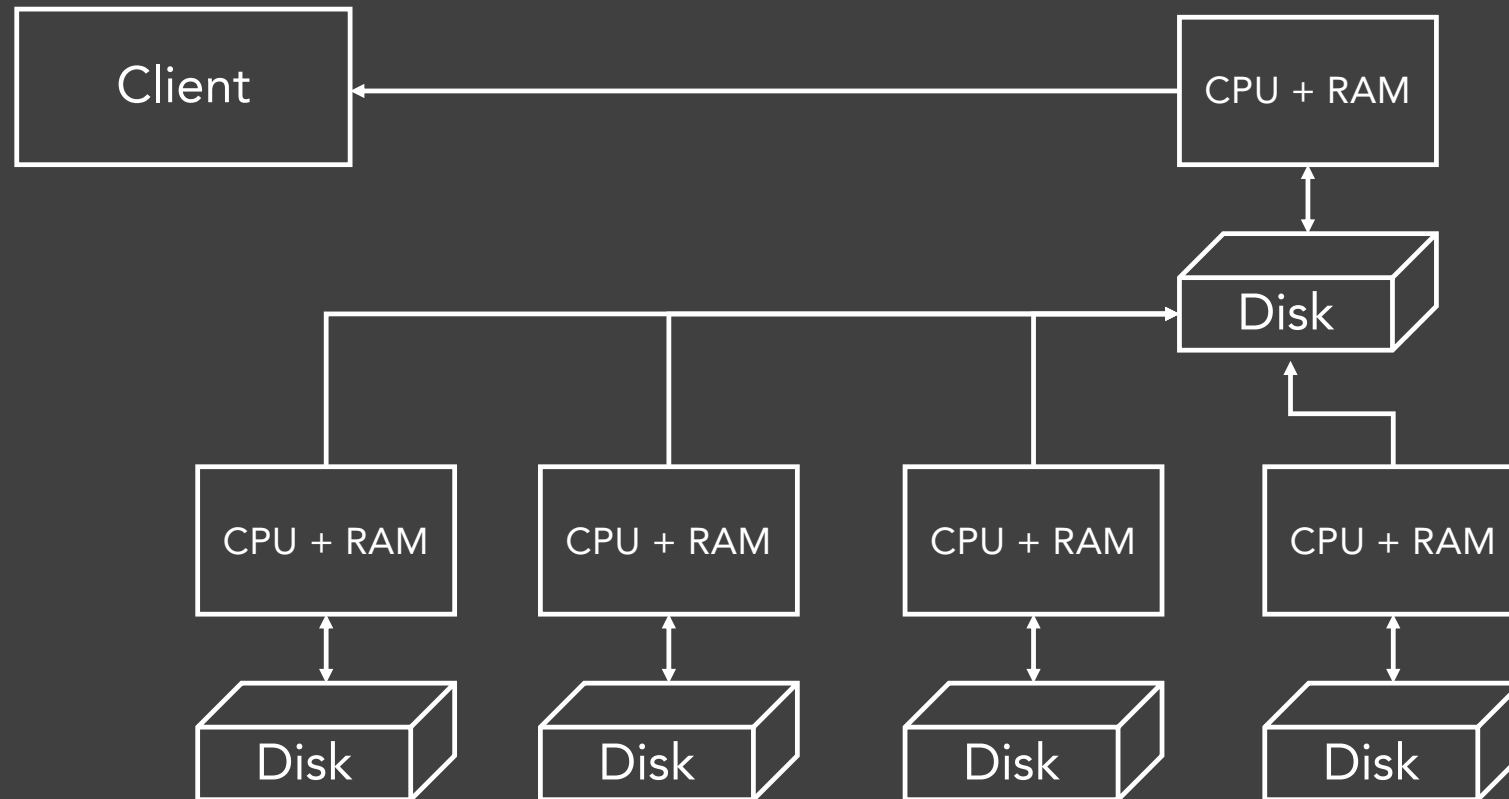
- Creates partitions for each reducer.
- No. of partitions = No. of reducer tasks.
- Same key will land on the same partition.
- Two types:
 - Default partitioner – Uses hash function to put records in partitions.
 - Custom partitioner – User created.

Combiner

- Mini reducer.
- Run at map side on mapper outputs.
- Reduces the output on mapper that have to be moved by generating aggregated output on each map output.



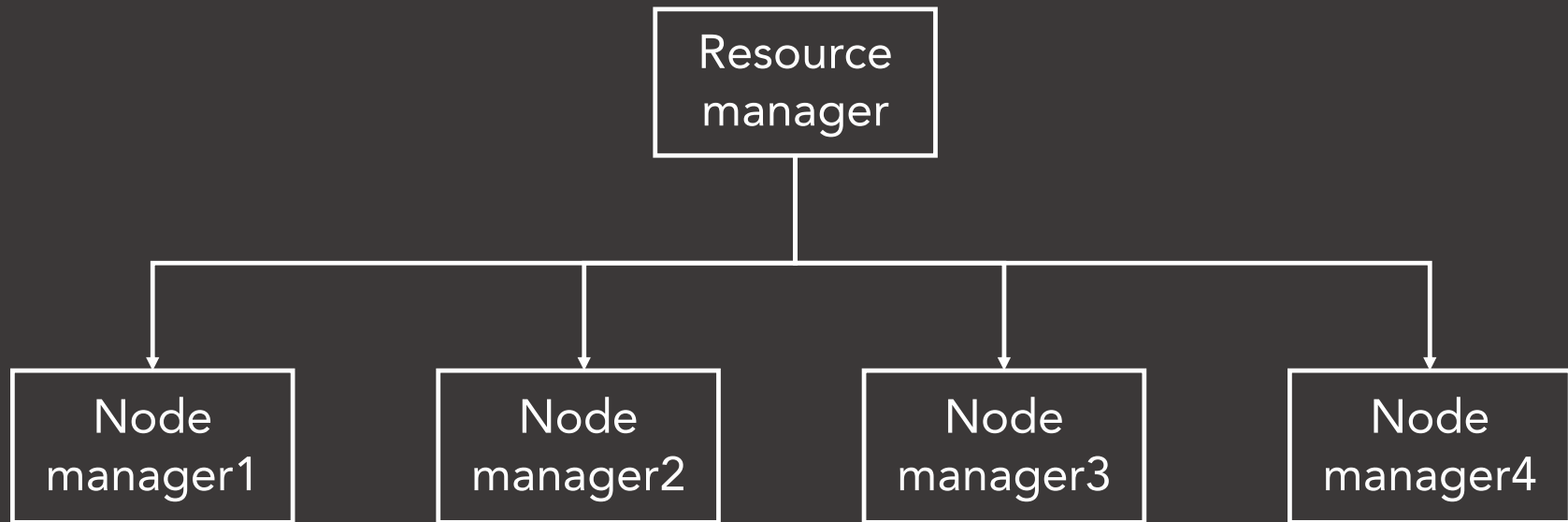
Distributed Processing



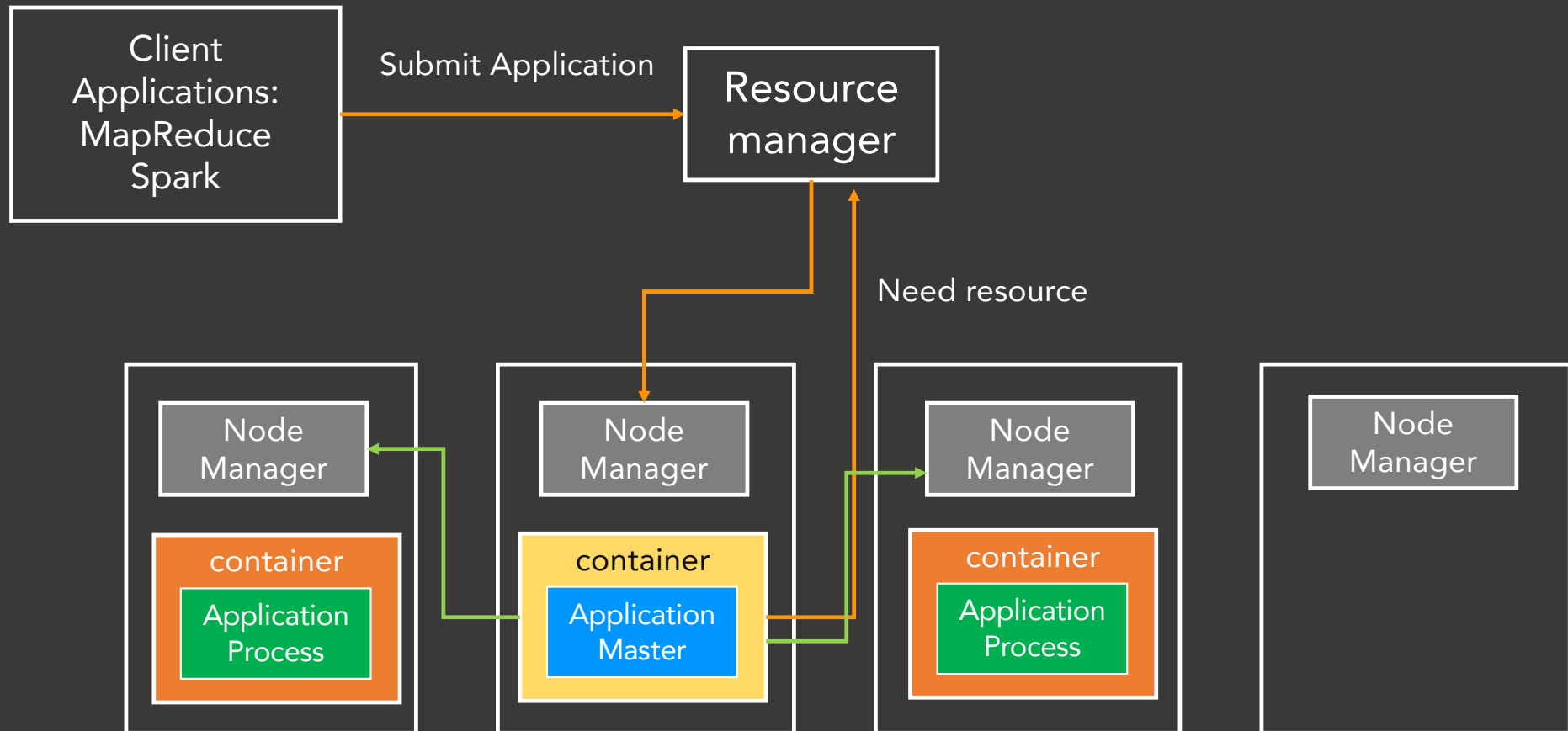
YARN

- Cluster resource manager.
- Have APIs to work with cluster resources.

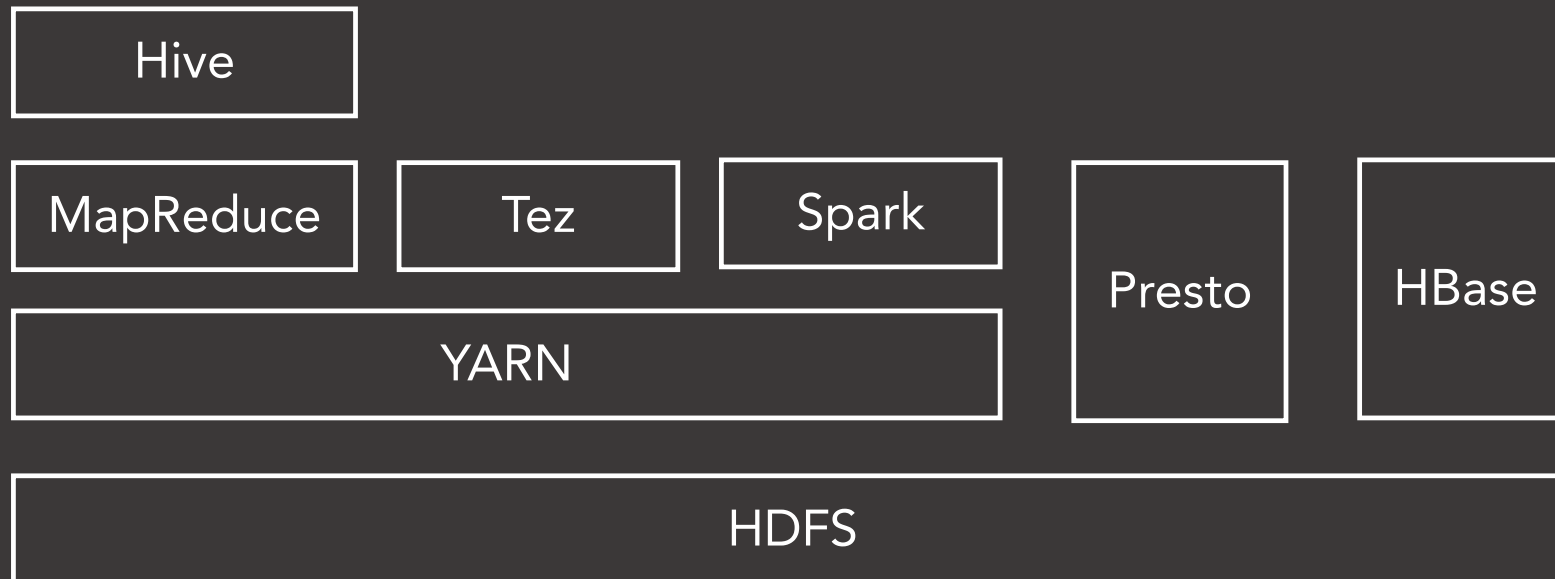
YARN



YARN



Hadoop Ecosystem



Thank You!